Do wide binaries follow a scale set by the cosmic expansion? A registered protocol for Gaia DR4, with a dry run on DR3

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Abstract

This record registers, before Gaia DR4 (planned for 2 December 2026), a test of wide binaries. It is one of the checks of Mathematical Materiality (MM): the outcome concerns MM's reading of wide binaries, not the framework. Two MM readings are compared. The main one is a chain of correspondences that takes no parameter from the binaries. It links three things that seem unrelated: The field equation (QUMOND) is borrowed. The declared alternative is the reading of MM 2.54, which says that wide binaries are Newtonian. MOND with the external field is reported as a reference. The data and the rules are fixed now. The pairs of El-Badry, Rix and Heintz (2021) will be re-measured in DR4, with the cuts frozen on 11 October 2026. Stars with a DR4 non-single-star solution are removed. A forward simulation models hidden companions. The decision rule compares the chain with Newton and has written validity conditions: the test can favour one reading, or come out inconclusive or not valid.

Version 2.56 of the Mathematical Materiality (MM) series, October 2026. A registration record: the protocol, code, sample and dry run, deposited before Gaia DR4 (planned for 2 December 2026). The protocol text of reference is documenti/PROTOCOLLO_DR4_v2.md (in Italian); this README summarises it. It refers to MM 2.54 (DOI 10.5281/zenodo.23418156), whose census lists wide binaries among the entries with a future check.

This is one of MM's checks. The test can favour one of two MM readings, or come out inconclusive or not valid. Whatever the outcome, it concerns the reading of wide binaries, not the framework.

1. What is compared

hypothesis	content	role
chain, $H_0 = 67.4$	$a_0 = cH_0/6 = 1.091 \cdot 10^{-10}$ m s^{-2} (the two-mirror reading of MM 2.54); the reading-series law $\nu(y) = 1/(1 - e^{\wedge}(-\sqrt{y}))$ (the MLS form of the radial acceleration relation); the Galactic external field from the rotation of the Milky Way (V = 233 km/s, R = 8.2 kpc: 1.966 a_0); QUMOND field equation with all multipoles (borrowed, not derived)	**MM, main reading**
chain, $H_0 = 73.04$	the same with $a_0 = 1.183 \cdot 10^{-10}$ (1.814 a_0)	sensitivity
Newton	Newtonian gravity	**MM, declared alternative** (the reading in MM 2.54)
MOND simple, MOND standard	QUMOND with the external field, $a_0 = 1.2 \cdot 10^{-10}$	reference
series, $a_0 = 1.2 \cdot 10^{-10}$	the reading-series law with the SPARC a_0 (earlier working reading)	reference

The chain takes no parameter from the binaries. It links three things that seem unrelated:

- the cosmic expansion, through $a_0 = cH_0/6$;
- the law read in galaxies;
- the rotation of the Milky Way.

Its asymptotic boost is $\eta = 1.327$ ($H_0 = 67.4$) and 1.362 ($H_0 = 73.04$).

2. Data

- **Pairs.** The wide binaries of El-Badry, Rix and Heintz 2021 (VizieR J/MNRAS/506/2269), downloaded on 16 October 2026 with $\varpi_1 > 4$ mas, sepAU > 200, R < 0.1: 167 301 pairs. The query and the SHA-256 are in dati/. The selected sample, with Gaia identifiers, is dati/campione_F.csv. With the frozen cuts it reproduces pair by pair the sample used in development (913 low-acceleration pairs, 21 489 control pairs).
- **DR4 values** (codice/registrazione/scarica_dr4.py). The same pairs, re-measured:
- through the DR3→DR4 neighbourhood table;
- or, if that table is missing, through a positional cross-match at J2017.5 within 0.3″;
- only one-to-one links are kept;
- stars with a DR4 non-single-star solution are removed. This is a new selection, not simulated.
- **Conversion** (converti_dr4.py). The DR4 values go into the format used in development, in canonical order and with SHA-256 fingerprints, and through the same chain of programs with the cuts frozen on 11 October 2026:
- BinType = MSMS;
- d < 200 pc;
- ϖ/σ_ϖ > 20;
- parallaxes consistent within 3σ;
- 3 < M_G < 12;
- error on u < 0.1;
- depth term < 0.1;
- u < 3.

3. Comparison and decision (exact rules in the protocol)

- **Forward simulation.**
- Hidden companions are modelled: astrometric fit, RUWE, El-Badry's cut, 2+2 systems and near-twins.
- Bands: a control band (4000 random pairs with x ≥ 10) and a low-acceleration band (x < 1).
- Grid: twins 4 % and 20 %, companions per star 0-0.4, 3 replicas.
- In all: 6 hypotheses, 60 files.
- **Likelihood.** Poisson on 26 equal-count classes per band. The twin fraction and the mass scale (0.85-1.15) are common to both bands; the companion fraction is separate for each band.
- **Validity.** The test is not valid if any of these fails:
- V1: at least 600 low-band and 15 000 control pairs;
- V2: libraries of 200 000 orbits with energy error below $2\cdot10^{-2}$ at the 99th percentile; no unbound initial state; complete grid; data fingerprints equal;
- V4: the best hypothesis has deviance ≤ 3 × the number of classes in each band.
- **Decision.** D = L(chain 67.4) − L(Newton); D_k is the same with one replica removed.
- **Chain favoured:** D ≥ +5 and every D_k ≥ +2.
- **Newton favoured:** D ≤ −5 and every D_k ≤ −2.
- **Otherwise:** inconclusive.
- Also reported: E = L(chain) − L(best MOND). The chain is «close to the best MOND» if E ≥ −5.

4. Dry run on DR3 (data already seen during development)

- **Same data, same simulations.** The DR3 values, written in the archive format and converted, give the same pairs and the same quantities as the development files (difference 0). With the same row order, the simulations rerun for Newton and for the chain are identical to the existing ones.
- **The rules applied to the 60 existing DR3 simulations:**

hypothesis	ΔlogL from the best	deviance control / low
MOND simple	0	43 / 21
MOND standard	−0.8	28 / 38
chain, H_0 = 67.4	**−2.4**	38 / 31
chain, H_0 = 73.04	−4.2	45 / 28
series, $a_0 = 1.2\cdot10^{-10}$	−8.7	49 / 33
Newton	**−10.0**	37 / 47

- **The numbers.** Valid; D = +7.64 (subsets +7.56, +4.50, +2.07); E = −2.36.
- **The decision on DR3** would be «chain favoured over Newton», with the minimum margin only just met.
- **The DR3 → DR4 download** was tried on DR3, both by identifier and by position (risultati/scarico/).
- **Not tested before DR4:** the DR4 neighbourhood table itself, the effect of the non-single-star selection, and a full rerun of the grid on new data.

5. Grade and declared limits

- **Grade B*.** The data were partly seen before registration (DR3), and DR4 reuses DR3 stars and observations. A favourable outcome is a confirmation on non-independent data, not a blind prediction.
- **Limits:**
- generator not rewritten from scratch; tails at 3σ not studied;
- masses with a free common scale, not full distributions;
- eccentricity fixed to Hwang et al. 2022;
- QUMOND borrowed;
- Galactic V and R fixed;
- non-single-star selection not simulated;
- pmRA-pmDE correlation not used.

The scope was reduced on purpose: this is one check among many.

- **Amendments.** Any change after deposit is a dated addendum, with its reason and the outcome under both the original and the new rule.

6. Verification

- **Two rounds by Codex/GPT-6** (documenti/verifica_astra/).
- First round, on version 1: «not ready», nine points, all addressed in version 2. The list is at the end of the protocol.
- Second round, on version 2: eight points resolved, no new error. One point was partly open: an explicit check of unbound Newtonian orbits, and a clean «not valid» on an incomplete grid instead of a crash.
- **After the second round** (valuta_dr4.py 2.1):
- the Newtonian library is checked orbit by orbit: 0 unbound states out of 200 000;
- an incomplete grid now gives «not valid», tested with a missing file and with a file of two replicas (risultati/prova_a_secco/prova_valutazione_casi.log).
- No third round.
- Every file was registered with SHA-256 and UTC time before it was run (documenti/registri/).

7. Files and reproducibility

- `codice/registrazione/`: download, conversion, simulation, evaluation and dry-run programs (version 2; version 1 in versione_1/).
- `codice/pipeline/`: the programs of the development chain they call, unchanged. The code contains the author's paths (E:\...), to be adapted.
- `dati/`: the query, the SHA-256 of the download, the selected sample with Gaia identifiers.
- `risultati/`:
- libraries: dynamic checks and SHA-256. The six libraries (217 MB each) are not included; they can be regenerated with orbite.py or orbite_mm.py and the seeds 1, 12, 13, 14, 15, 16 (FATT_PASSI = 1 for Newton, 4 for the others);
- the sample, the chain, the dry run, the download tests.
- `documenti/`: the plan, the decision on the main reading, the protocol versions, the verification report, the registers.

The author is not a professional physicist; computations were done with AI assistants (Claude, with two checks by Codex), not reviewed by a human referee. Texts and materials: CC BY 4.0; code: MIT.

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